

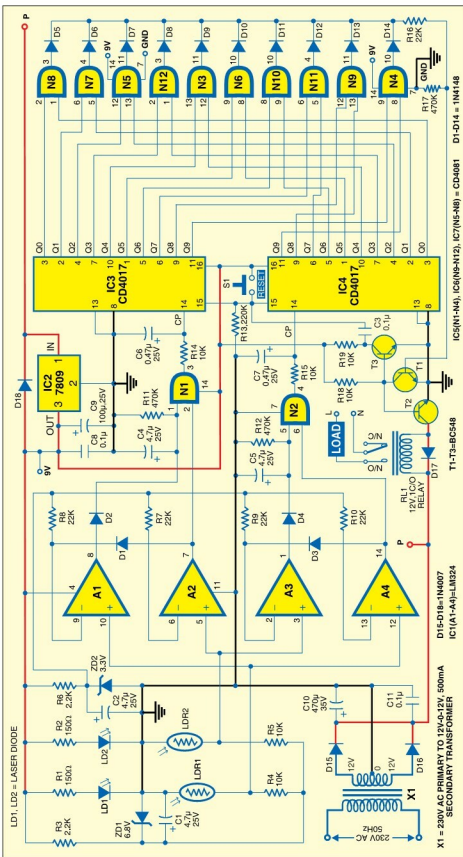


Fig. 1: Circuit of automatic room power controller

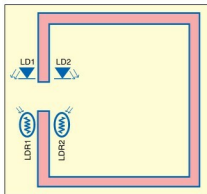


Fig. 2: Sensor arrangement on the door

Inverting input pins 2, 6, 9 and 13 are held at a constant voltage of 3.3V derived from the combination of resistor R6 and zener diode ZD2 (3.3V zener diode).

Suppose LDR1 is fitted outside the door and LDR2 inside (reverse side). When anyone enters the room light on LDR1 is blocked first (its resistance increases) and pins 10 and 12 of IC1 are at higher potential than its pins 9 and 13. The high output from pins 8 and 14 of IC1 is fed to N1 and N2 respectively. Capacitor C4 connected to N1 charges and retains pin 1 of N1 high for a few seconds before it discharges through resistor R11.

If light on LDR2 is blocked within this time period, pins 5 and 3 of IC1 become high. The high output at pin 7 of operational amplifier A2 is fed to N1, which provides a clock pulse at pin 14 of IC3. Pins 1 and 14 of IC1 cannot become high together because the high output at pin 14 makes the voltage at pin 2 higher than pin 3 (via diode D3) and pin 1 becomes low. Thus pins 5 and 6 of N2 don't go high at the same time and IC4 gets no clock pulse when someone enters the room.

When someone comes out of the room, LDR2 is blocked first making pins 7 and 1 of IC1 high. The high status of pin 1 is retained for a few seconds by capacitor C5 (via diode D4) until it discharges through resistor R12.

If light falling on LDR1 is